

Generated by Scopus AI, Fri Mar 27 2026

Hydrology studies on tropical coastal ecosystems

Quantitative Hydrological Comparison of Urban Compacted Soils and Mangrove/Natural Vegetation in Tropical Coastal Ecosystems

Quick Reference
Key Findings Table

Parameter	Urban Compacted Soils / Parks	Mangrove / Dense Native Vegetation	DOI / Reference(s)
Infiltration Rate (mm/h)	8–175 (often <20 in SE Asia); as low as 0.3–6.9 (compacted sandy); up to 35.8 (urban gardens)	225–652 (non-compacted, vegetated); up to 470 (urban park, dense vegetation); 81 (tropical forest); 121.9 (riparian forest)	1 2 3 4 5
Runoff Coefficient (%)	0.77 (77%) in settlements; 0.245 in urban parks (vegetation); up to 1.0 (impervious)	0.35 (35%) in forests/mangroves; ~0.25 in dense vegetation	2 6
Peak Flood Reduction	Not directly reported for urban parks	Attenuation: 42 ± 9.8 to 46 ± 9.8 cm/km; Storm surge height reduction: 4–16.5 cm; Water flow velocity reduction: 29–92%	7 8 9

Direct Answer

- **Infiltration rates** in compacted urban soils are typically **8–175 mm/h** (often <20 mm/h in Southeast Asia), while **natural/mangrove soils** can reach **225–652 mm/h**.
- **Runoff coefficients** are **77%** in urban settlements versus **35%** in forested/mangrove areas.
- **Peak flood reduction** in mangroves: **attenuation rates of 42 ± 9.8 to 46 ± 9.8 cm/km**, and **storm surge height reductions of 4–16.5 cm**.
- **DOIs/References:** [1](#) [2](#) [4](#) [6](#) [7](#) [8](#) [9](#)

Unable to Find

- No studies provide direct, side-by-side quantitative averages (with DOIs) for **Rhizophora mangrove soils vs. compacted urban soils** in tropical Latin America. Most data are from Southeast Asia or are not paired comparisons.

Study Scope

- **Time Period:** Primarily post-2010, with emphasis on studies after 2015.

- **Disciplines:** Hydrology, soil science, urban ecology, coastal ecosystem management.
- **Methods:** Field infiltration tests, runoff hydrograph analysis, remote sensing, hydrological modeling (SWAT, Curve Number, Horton, Green-Ampt), AI/machine learning.

Assumptions & Limitations

- Most studies compare urban and natural/mangrove soils in different catchments, not always side-by-side.
- Soil texture, vegetation type, and management practices vary widely, affecting comparability.
- Data for Latin America are sparse; Southeast Asia is better represented.
- Peak flood reduction is often modeled or inferred, not always measured directly.

Suggested Further Research

- Standardized, paired field studies in underrepresented regions (especially Latin America).
- Integration of high-resolution remote sensing and AI for spatially explicit hydrological monitoring.
- Improved models that incorporate mangrove-specific processes and urban landscape metrics.

1. Introduction

Context and Rationale

Tropical coastal zones are increasingly urbanized, leading to widespread soil compaction, loss of native vegetation, and increased flood risk. Urban expansion replaces permeable, vegetated soils with impervious surfaces, drastically altering hydrological processes—most notably by reducing infiltration and increasing surface runoff and flood peaks. In contrast, mangrove forests (e.g., *Rhizophora* spp.) and dense native vegetation provide critical ecosystem services, including flood attenuation, storm surge mitigation, and enhanced infiltration. Quantitative comparisons of these scenarios are essential for evidence-based urban planning and ecosystem-based adaptation strategies, yet such data remain limited, especially in Latin America and for direct urban-mangrove contrasts [10](#) [11](#) [12](#).

2. Quantitative Comparison of Hydrological Parameters

2.1 Infiltration Rates: Urban Compacted Soils vs. Mangrove/Natural Vegetation

- **Urban Compacted Soils:**
 - Infiltration rates are severely reduced by compaction and loss of macropores.
 - Reported rates:
 - 8–175 mm/h (compacted sandy soils, Florida) [1](#)
 - <20 mm/h common in Southeast Asian urban soils [13](#)
 - 0.3–6.9 mm/h (compacted sandy, urban) [14](#)

- 35.8 mm/h (urban gardens, Brazil) [5](#)
- Soil amendments (e.g., tillage, compost) can increase rates to 20–35 cm/h (200–350 mm/h) over several years [15](#) [16](#)

- **Mangrove/Natural Vegetation Soils:**

- Infiltration rates are much higher due to organic matter, root structure, and less compaction.
- Reported rates:
 - 225–652 mm/h (non-compacted, vegetated soils, Florida) [1](#)
 - 81 mm/h (tropical forest, Sumani Watershed, Indonesia) [2](#)
 - 121.9 mm/h (riparian forest, Brazil) [5](#)
 - 470 mm/h (urban park, dense native vegetation, Australia) [4](#)
 - Mangrove soils: high porosity and organic content, but exact mm/h values for *Rhizophora* are rarely reported [17](#) [18](#)

Synthesis:

Urban compaction reduces infiltration by 70–99% compared to natural/mangrove soils. Soil amendments can partially restore infiltration, but not to mangrove levels. Vegetation type and soil texture (sandy > loamy > clayey) are key determinants [1](#) [14](#) [19](#).

2.2 Runoff Coefficients and Surface Runoff

- **Urban Compacted Soils:**

- Runoff coefficients are high due to low infiltration and impervious surfaces.
- 0.77 (77%) in urban settlements (Sumani Watershed, Indonesia) [2](#)
- 0.245 (24.5%) for vegetated areas in urban parks; up to 1.0 for impervious surfaces [6](#)
- Urbanization increases runoff and decreases infiltration [11](#) [20](#)

- **Mangrove/Dense Native Vegetation:**

- Runoff coefficients are much lower.
- 0.35 (35%) in forested/mangrove areas [2](#)
- ~0.25 in dense native vegetation in urban parks [6](#)
- Sustainable land management (e.g., reforestation) can reduce runoff by ~30% [2](#)

Synthesis:

Dense vegetation and mangroves reduce runoff coefficients by 30–50% compared to urban soils. Soil texture,

vegetation cover, and management practices modulate these effects [2](#) [6](#).

2.3 Peak Flood Reduction and Attenuation

- **Mangrove Forests (especially *Rhizophora*):**

- Attenuation rates: 42 ± 9.8 to 46 ± 9.8 cm/km (highest recorded for mangroves) [7](#)
- Storm surge height reduction: 4–16.5 cm (Bangladesh, depending on forest width/density) [8](#)
- Water flow velocity reduction: 29–92% [8](#)
- Field measurements: up to 9.4 cm/km reduction in peak water level during hurricanes (Florida) [9](#)
- Mangrove widths >500 m dissipate 75%+ of wave energy [21](#)

- **Urban Parks:**

- No direct data on peak flood reduction in tropical urban parks; constructed wetlands in temperate zones reduce pollutant loads but not directly comparable [22](#)

Synthesis:

Mangroves, especially *Rhizophora*, provide substantial flood attenuation, reducing peak water levels and storm surge heights. Urban parks offer some hydrological benefits but are less effective than intact mangrove forests

[7](#) [8](#) [9](#).

3. Categorization of Data by Soil, Vegetation Type, and Geographic Location

3.1 Soil Texture and Vegetation Influence

- **Sandy Soils:**

- Higher infiltration (e.g., loamy sand: 3–4 cm/h; sandy loam: 1.8–2.3 cm/h) [23](#)
- Compaction reduces rates to 0.3–6.9 mm/h [14](#)
- Mangrove species distribution correlates with soil texture (e.g., *Bruguiera* in sandy, *Ceriops* in clay/silt) [24](#)

- **Clayey Soils:**

- Lower infiltration (0.25–0.58 cm/h), but high organic content in mangroves can improve structure [18](#) [23](#)
- Compaction and moisture content both critical [1](#) [25](#)

- **Vegetation Type:**

- Dense native vegetation and mangroves maintain lower bulk density, higher porosity, and better infiltration [4](#) [26](#)

- Above-ground roots (e.g., *Rhizophora* prop roots) enhance flow attenuation [27](#)

Synthesis:

Soil texture and vegetation type interact to determine infiltration and runoff. Mangroves and dense native vegetation consistently outperform compacted urban soils, especially in sandy or loamy contexts [4](#) [14](#) [24](#).

3.2 Geographic Variability: Latin America vs. Southeast Asia

• Latin America:

- Forested areas: infiltration up to 86 cm/h (Colombia); mixed/compacted: as low as 4 cm/h [2](#) [28](#)
- Urban parks: rapid infiltration (470 mm/h, Australia; similar trends in Brazil) [4](#) [5](#)
- Runoff coefficients higher in urban areas, lower in forests [29](#) [30](#)

• Southeast Asia:

- Urban soils: often <20 mm/h infiltration [13](#)
- Forested/mangrove: up to 81 mm/h (Sumani Watershed, Indonesia) [2](#)
- Runoff coefficients: 77% (urban), 35% (forest) [2](#)

Synthesis:

Both regions show similar trends: urbanization reduces infiltration and increases runoff. Southeast Asia provides more paired data; Latin America is underrepresented [2](#) [28](#).

3.3 Urban Parks and Dense Native Vegetation

• Infiltration Rates:

- Dense native vegetation: up to 470 mm/h [4](#)
- Compacted/reconstructed areas: much lower rates [4](#) [26](#)

• Runoff Coefficients:

- Vegetation areas: ~0.245 [6](#)
- Impervious: up to 1.0 [6](#)

• Hydrological Benefits:

- Dense vegetation reduces runoff, increases groundwater recharge (up to 11.8% of rainfall) [6](#)
- Soil compaction impairs infiltration and root growth [4](#) [26](#)

Synthesis:

Urban parks with dense native vegetation can approach the hydrological performance of natural systems, but compaction and disturbance reduce these benefits [4](#) [6](#).

4. Methodological Approaches in Hydrological Assessment

4.1 Field Measurement Techniques

- **Infiltration:**

- Double-ring infiltrometers, rainfall simulators, automated infiltrometers (e.g., SATURO) [23](#) [31](#)
- Soil property assessments: bulk density, porosity, organic content [10](#) [32](#)

- **Runoff:**

- Hydrograph analysis, plot-scale runoff modeling, Curve Number (CN) method [33](#) [34](#)
- Water level loggers, storm event monitoring [35](#)

- **Flood Attenuation:**

- High water level attenuation measured in mangroves during storm events [7](#) [9](#)

Synthesis:

Field methods provide robust, site-specific data but are labor-intensive and often limited in spatial/temporal coverage [31](#) [35](#).

4.2 Hydrological Modeling Approaches

- **Models Used:**

- SWAT, Curve Number (CN), Horton, Green-Ampt, Kostiakov, Philip [33](#) [36](#) [37](#)
- Unit hydrograph, kinematic wave routing [35](#)

- **Strengths:**

- Allow scenario analysis, integration of land use/soil data, and prediction under different rainfall events [38](#) [39](#)

- **Limitations:**

- SWAT/CN often underestimate runoff at low rainfall, require extensive data, and do not explicitly model mangrove-specific processes (e.g., tidal effects, root structure) [36](#) [40](#) [41](#)
- Calibration is challenging in data-scarce regions [42](#)

Synthesis:

Models are essential for scaling up findings but need adaptation for tropical, mangrove-rich, and urbanized contexts [36](#) [41](#).

4.3 Remote Sensing and AI-Based Methods

- **Applications:**

- Mapping infiltration capacity, runoff, and mangrove extent using multispectral imagery, SAR, and machine learning (random forest, XGBoost) [43](#) [44](#)
- AI models (e.g., random forest, neural networks) predict infiltration rates and flood volume reduction [45](#) [46](#)
- Overcome cloud cover limitations with SAR [47](#)

- **Advances:**

- High-resolution, spatially explicit monitoring; integration with field data [48](#) [49](#)
- Improved accuracy in heterogeneous urban/mangrove landscapes [44](#) [50](#)

- **Challenges:**

- Data integration, model interpretability, and need for ground-truthing [51](#) [52](#)

Synthesis:

Remote sensing and AI are rapidly improving hydrological monitoring, especially in data-poor tropical regions [44](#) [52](#).

5. Critical Analysis of Findings

- **Consistency:**

- All lines of evidence confirm that urban compaction drastically reduces infiltration and increases runoff, while mangroves and dense native vegetation provide substantial hydrological benefits [1](#) [2](#) [6](#).

- **Variability:**

- Soil texture, vegetation type, and management practices cause wide variability in measured rates, emphasizing the need for local data [14](#) [19](#).

- **Data Gaps:**

- Direct, paired comparisons (especially with DOIs) are rare, particularly for *Rhizophora* mangroves vs. urban soils in Latin America [18](#) [53](#).

- **Methodological Limitations:**

- Models often lack mangrove-specific processes and require extensive calibration [36](#) [41](#).

- **Emerging Solutions:**

- Remote sensing and AI offer promise for bridging data gaps and improving spatial/temporal resolution [44](#) [52](#).

6. Real-world Implications

- **Urban Planning:**

- Maintaining or restoring mangroves and dense native vegetation in coastal cities can significantly reduce flood risk and surface runoff, providing cost-effective ecosystem-based adaptation [7](#).

- **Ecosystem Services:**

- Mangroves offer superior flood attenuation compared to conventional urban parks, justifying their conservation and integration into urban resilience strategies [8](#) [9](#).

- **Stormwater Management:**

- Soil amendments and green infrastructure can partially restore infiltration in urban areas but are not substitutes for intact mangrove systems [15](#) [16](#).

- **Policy:**

- Quantitative evidence supports prioritizing mangrove conservation and restoration in coastal urban planning [7](#).

7. Future Research Directions

Data Gaps in Quantitative Hydrological Studies

- **Needs:**

- Paired, standardized field studies comparing urban and mangrove soils, especially in Latin America and Africa [12](#) [54](#).
- High-resolution, long-term monitoring of infiltration, runoff, and flood attenuation [55](#) [56](#).

Limitations of Current Hydrological Models

- **Improvements:**

- Incorporate mangrove-specific processes (e.g., root structure, tidal effects) into models [41](#).
- Enhance calibration with local field data and remote sensing [42](#) [57](#).

Recommendations for Future Research

- **Methodological:**

- Integrate field measurements, remote sensing, and AI for robust, spatially explicit hydrological assessments [45](#) [52](#).
- Develop and validate models tailored to tropical urban-mangrove systems [41](#).

- **Geographic:**

- Expand research to underrepresented regions and small catchments [12](#) [58](#).

- **Socio-ecological:**

- Link hydrological data with socio-economic and land-use change analyses for comprehensive flood risk management [59](#) [60](#).

Synthesis

A robust body of evidence demonstrates that urban compaction in tropical coastal zones severely impairs infiltration and increases runoff, while mangroves and dense native vegetation provide substantial flood mitigation through higher infiltration rates and significant peak flow attenuation. However, direct, DOI-linked quantitative comparisons remain rare, especially for *Rhizophora* mangroves versus urban soils in Latin America. Methodological advances in remote sensing and AI are beginning to bridge these gaps, but further standardized, multidisciplinary research is needed to fully quantify and harness the ecosystem services of mangroves for urban flood resilience.

References (DOI/Identifiers):

[1](#) [2](#) [4](#) [6](#) [7](#) [8](#) [9](#) [14](#) [18](#) [23](#) [31](#) [36](#) [41](#) [44](#) [52](#)

References

1. Effect of urban soil compaction on infiltration rate Gregory, J.H., Dukes, M.D., Jones, P.H., Miller, G.L. Journal of Soil and Water Conservation, 2006 <https://www.scopus.com/pages/publications/33646843131?origin=scopusAI>
2. THE ROLE OF LAND USE COMPOSITION ON SURFACE FLOW UPSTREAM OF THE SUMANI WATERSHED Aprisal, Lusimaira, Yasin, S., (...), Putri, S.P. Water Conservation and Management, 2025 <https://www.scopus.com/pages/publications/105003597556?origin=scopusAI>
3. Basic infiltration rate and granulometric characteristics of an urban area in the Brazilian savannah Monteiro, V.V., Da Silva, S., Santos, F.C.V.D., De Sousa, A.C. Revista em Agronegocio e Meio Ambiente, 2021 <https://www.scopus.com/pages/publications/85122536691?origin=scopusAI>
4. Naturalization as a strategy for improving soil physical characteristics in a forested urban park Millward, A.A., Paudel, K., Briggs, S.E. Urban Ecosystems, 2011 <https://www.scopus.com/pages/publications/79955075212?origin=scopusAI>
5. Urban Gardens' Potential to Improve Stormwater Management: A Comparative Analysis among Urban Soils in Sorocaba, São Paulo, Brazil Corrêa, C.J.P., Nnadi, E.O., Mbanaso, F.U., Tonello, K.C. Sustainability (Switzerland), 2022 <https://www.scopus.com/pages/publications/85126323399?origin=scopusAI>
6. Evaluation and Improvement Measures of the Runoff Coefficient of Urban Parks for Sustainable Water Balance Son, J., Kwon, T. Land, 2022 <https://www.scopus.com/pages/publications/85137318352?origin=scopusAI>
7. Strong attenuation of high water levels observed in tropical mangroves Pelckmans, I., Vermeulen, B., Ramos-Veliz, J.A., (...), Temmerman, S. Limnology and Oceanography, 2025 <https://www.scopus.com/pages/publications/105010614465?origin=scopusAI>

8. Quantifying the protective capacity of mangroves from storm surges in coastal Bangladesh Dasgupta, S., Islam, M.S., Huq, M., (...), Hasib, M.R. PLoS ONE, 2019
<https://www.scopus.com/pages/publications/85063347330?origin=scopusAI>
9. Water level observations in mangrove swamps during two hurricanes in Florida Krauss, K.W., Doyle, T.W., Doyle, T.J., (...), Conner, W.H. Wetlands, 2009 <https://www.scopus.com/pages/publications/77749292977?origin=scopusAI>
10. Water entry, retention and movement in soils of warm humid regions Rahman, A. Hydrology of warm humid regions. Proc. international symposium, Yokohama, 1993, 1993
<https://www.scopus.com/pages/publications/0027802368?origin=scopusAI>
11. Changes of rainfall infiltration and runoff process due to urbanization Oka, T. null, 1993
<https://www.scopus.com/pages/publications/85058870010?origin=scopusAI>
12. Impacts of urbanization on stormflow magnitudes in small catchments in the Sandhills of South Carolina, USA Hung, C.-L.J., James, L.A., Carbone, G.J. Anthropocene, 2018
<https://www.scopus.com/pages/publications/85051392362?origin=scopusAI>
13. The Determination of Soil Infiltration Rate of Urban Bioretention Design Process in Chiang Mai, Thailand Sittisom, P., Tangsongsuwan, R., Munlikawong, S., (...), Rinchumphu, D. Nakhara: Journal of Environmental Design and Planning, 2022 <https://www.scopus.com/pages/publications/85147434129?origin=scopusAI>
14. Measurements of infiltration rates in compacted urban soils Pitt, R., Chen, S.-E., Ong, C.K., Clark, S. Proceedings of the Engineering Foundation Conference, 2001
<https://www.scopus.com/pages/publications/62649117318?origin=scopusAI>
15. Amending soils for enhanced infiltration of stormwater Kays, B.L., McLaughlin, R., Heitman, J., (...), Brown, V. International Low Impact Development Conference 2015 - LID: It Works in All Climates and Soils - Proceedings of the 2015 International Low Impact Development Conference, 2015
<https://www.scopus.com/pages/publications/84922879855?origin=scopusAI>
16. Effect of amendment type and incorporation depth on runoff from compacted sandy soils Bean, E.Z., Dukes, M.D. Journal of Irrigation and Drainage Engineering, 2015
<https://www.scopus.com/pages/publications/84929754137?origin=scopusAI>
17. Determination of photon shielding parameters for soils in mangrove forests Hila, F.C., Dicen, G.P., Javier-Hila, A.M.V., (...), Amorsolo, A.V. Philippine Journal of Science, 2021
<https://www.scopus.com/pages/publications/85100373365?origin=scopusAI>
18. Soil organic carbon of mangrove forests (Rhizophora and Avicennia) of the Venezuelan Caribbean coast Barreto, M.B., Lo Mónaco, S., Díaz, R., (...), Peralba, M.D.C.R. Organic Geochemistry, 2016
<https://www.scopus.com/pages/publications/84983507798?origin=scopusAI>
19. Soil texture mediates the surface cooling effect of urban and peri-urban green spaces during a drought period in the city area of Hamburg (Germany) Stumpe, B., Bechtel, B., Heil, J., (...), Marschner, B. Science of the Total Environment, 2023 <https://www.scopus.com/pages/publications/85165527959?origin=scopusAI>
20. Urbanization and Its Effect on Storm Water Management Ighare, S.V., Pradhan, V.S. Lecture Notes in Civil Engineering, 2025 <https://www.scopus.com/pages/publications/85211594721?origin=scopusAI>
21. Quantifying uncertainty in wave attenuation by mangroves to inform coastal green belt policies van Wesenbeeck, B.K., van Zelst, V.T.M., Antolinez, J.A.A., de Boer, W.P. Communications Earth and Environment, 2025 <https://www.scopus.com/pages/publications/105002763227?origin=scopusAI>
22. Evaluating the Effectiveness of Two Stormwater Wetlands in North Carolina Line, D.E., Jennings, G.D., Shaffer, M.B., (...), Hunt, W.F. American Society of Agricultural and Biological Engineers Annual International Meeting 2008, ASABE 2008, 2008 <https://www.scopus.com/pages/publications/63049133922?origin=scopusAI>

23. Rate of infiltration for different soil textures using rainfall simulator and Green–Ampt model Bharali, B. ISH Journal of Hydraulic Engineering, 2021 <https://www.scopus.com/pages/publications/85102939659?origin=scopusAI>
24. Mangrove species distribution across soil texture gradients in Benoa Bay and Nusa Lembongan, Bali Province, Indonesia Dharmayasa, I.G.N.P., Sugiana, I.P., Anantanasakul, P. Biodiversitas, 2025 <https://www.scopus.com/pages/publications/105011171700?origin=scopusAI>
25. Compaction's impacts on urban storm-water infiltration Pitt, R., Chen, S.-E., Clark, S.E., (...), Ong, C.K. Journal of Irrigation and Drainage Engineering, 2008 <https://www.scopus.com/pages/publications/52649097503?origin=scopusAI>
26. Impacts of human activities on soil physical properties of urban green areas: A case study in thessaloniki city, Greece Matziris, E., Stefanou, S., Papazafeiriou, A.Z., (...), Papaioannou, A. Carpathian Journal of Earth and Environmental Sciences, 2016 <https://www.scopus.com/pages/publications/84964743030?origin=scopusAI>
27. Measurement and Modeling of Above-Ground Root Systems as Attributes of Flow and Wave Attenuation Function of Mangroves Yoshikai, M., Nakamura, T., Suwa, R., (...), Nadaoka, K. Mangroves: Ecology, Biodiversity and Management, 2021 <https://www.scopus.com/pages/publications/85163484052?origin=scopusAI>
28. Variability of soil surface characteristics in a mountainous watershed in Valle del Cauca, Colombia: Implications for runoff, erosion, and conservation Guzman, C.D., Hoyos-Villada, F., Da Silva, M., (...), Steenhuis, T.S. Journal of Hydrology, 2019 <https://www.scopus.com/pages/publications/85067669308?origin=scopusAI>
29. Runoff Coefficients Associated with Different Return Periods Calculated by Means of Statistical Analysis Ambriz, M.G.R., Mora, R.D., Juarez, M.L.A., Elizondo, E.C. Proceedings of the IAHR World Congress, 2022 <https://www.scopus.com/pages/publications/85178365758?origin=scopusAI>
30. Morphometric Characterization of the Sierra de Manantlán Biosphere Reserve Water Producing Area Guevara-Gutiérrez, R.D., Barreto-Garcia, O.A., Olguín-López, J.L., (...), Fletes, J.A.L. Investigaciones Geograficas, 2025 <https://www.scopus.com/pages/publications/105013751799?origin=scopusAI>
31. Temporal and spatial variation of infiltration in urban green infrastructure Ebrahimian, A., Sample-Lord, K., Wadzuk, B., Traver, R. Hydrological Processes, 2020 <https://www.scopus.com/pages/publications/85075259902?origin=scopusAI>
32. Comprehensive evaluation of soil characteristics and carbon stocks variability in different urban land use types Reyes, N.J.D., Geronimo, F.K.F., Choi, H., (...), Kim, L.-H. Environmental Engineering Research, 2024 <https://www.scopus.com/pages/publications/85185668542?origin=scopusAI>
33. Investigation of the Curve Number Method For Surface Runoff Estimation In Tropical Regions Dile, Y.T., Karlberg, L., Srinivasan, R., Rockström, J. Journal of the American Water Resources Association, 2016 <https://www.scopus.com/pages/publications/84989225515?origin=scopusAI>
34. Plot-scale rainfall-runoff characteristics and modeling at six sites in Australia and Southeast Asia Yu, B., Rose, C.W., Coughlan, K.J., Fentie, B. Transactions of the American Society of Agricultural Engineers, 1997 <https://www.scopus.com/pages/publications/0031239346?origin=scopusAI>
35. Attenuation performance of runoff storage basins within a moderate to steep slope urban catchment in Cebu, Philippines Lalisán, C.L., Fornis, R.L. AIP Conference Proceedings, 2020 <https://www.scopus.com/pages/publications/85096500728?origin=scopusAI>
36. Development and application of a physically based landscape water balance in the SWAT model White, E.D., Easton, Z.M., Fuka, D.R., (...), Steenhuis, T.S. Hydrological Processes, 2011 <https://www.scopus.com/pages/publications/79952060497?origin=scopusAI>

37. Evaluation of infiltration models for mineral soils with different land uses in the tropics Suryoputro, N., Suhardjono, Soetopo, W., (...), Limantara, L.M. *Journal of Water and Land Development*, 2018
<https://www.scopus.com/pages/publications/85049230215?origin=scopusAI>
38. DEVELOPMENT AND ANALYSIS OF A GIS-BASED CURVE NUMBER MAP OF THE PHILIPPINES Alcober, C.A., Macuha, R. *ASEAN Engineering Journal*, 2024
<https://www.scopus.com/pages/publications/85194894704?origin=scopusAI>
39. An assessment of methods for estimating runoff rates at the plot scale Yu, B., Cakurs, U., Rose, C.W. *Transactions of the American Society of Agricultural Engineers*, 1998
<https://www.scopus.com/pages/publications/0032077812?origin=scopusAI>
40. Fields of Application of SWAT Hydrological Model—A Review Janjić, J., Tadić, L. *Earth (Switzerland)*, 2023 <https://www.scopus.com/pages/publications/85163734401?origin=scopusAI>
41. The effectiveness of mangrove forests as a nature-based solution against flood risk under an extreme weather event Shokatian-Beiragh, M., Banan-Dallalian, M., Golshani, A., (...), Samiee-Zenoozian, M. *Regional Studies in Marine Science*, 2024 <https://www.scopus.com/pages/publications/85196043606?origin=scopusAI>
42. Critical review of SWAT applications in the upper Nile basin countries Van Griensven, A., Ndomba, P., Yalew, S., Kilonzo, F. *Hydrology and Earth System Sciences*, 2012
<https://www.scopus.com/pages/publications/84869464525?origin=scopusAI>
43. An application of remote sensing in hydrology - A case study in Malaysia Wong, T.H., Mansor, S.B., Mispan, M.R., (...), Ahmad, N. *International Geoscience and Remote Sensing Symposium (IGARSS)*, 2001
<https://www.scopus.com/pages/publications/0035569110?origin=scopusAI>
44. Progress in quantitative remote sensing research of mangroves Wang, J., Li, Q., Wu, G. *National Remote Sensing Bulletin*, 2025 <https://www.scopus.com/pages/publications/105009535615?origin=scopusAI>
45. Assessment of the Experimental Rain Garden Infiltration Rate Using Artificial Intelligence Techniques Kumar, S., Singh, K.K. *World Environmental and Water Resources Congress 2024: Climate Change Impacts on the World We Live In - Proceedings of the World Environmental and Water Resources Congress 2024*, 2024
<https://www.scopus.com/pages/publications/85194416483?origin=scopusAI>
46. A new framework for urban flood volume estimation using low-impact development methods and intelligent models Dadrassajirlou, Y., Karami, H., Rezaei, A., (...), Ferreira, C.S.S. *Nature-Based Solutions in Supporting Sustainable Development Goals: Theory and Practice*, 2024
<https://www.scopus.com/pages/publications/85214152521?origin=scopusAI>
47. Mapping changes in the largest continuous Amazonian mangrove belt using object-based classification of multisensor satellite imagery Nascimento, W.R., Souza-Filho, P.W.M., Proisy, C., (...), Rosenqvist, A. *Estuarine, Coastal and Shelf Science*, 2013 <https://www.scopus.com/pages/publications/84873723664?origin=scopusAI>
48. Cumulative infiltration and infiltration rate prediction using optimized deep learning algorithms: A study in Western Iran Panahi, M., Khosravi, K., Ahmad, S., (...), Lee, C.-W. *Journal of Hydrology: Regional Studies*, 2021 <https://www.scopus.com/pages/publications/85105350635?origin=scopusAI>
49. A comparison of data mining techniques and multi-sensor analysis for inland marshes delineation Simioni, J.P.D., Guasselli, L.A., de Oliveira, G.G., (...), de Oliveira, G. *Wetlands Ecology and Management*, 2020
<https://www.scopus.com/pages/publications/85086164004?origin=scopusAI>
50. Soil infiltration rates are underestimated by models in an urban watershed in central North Carolina, USA Bergeson, C.B., Martin, K.L., Doll, B., Cutts, B.B. *Journal of Environmental Management*, 2022
<https://www.scopus.com/pages/publications/85127756398?origin=scopusAI>
51. Quantification of infiltration processes in urban areas by accounting for spatial parameter variability Mohrlök, U., Wolf, L., Klingner, J. *Journal of Soils and Sediments*, 2008
<https://www.scopus.com/pages/publications/40449121516?origin=scopusAI>

52. A CiteSpace-Based Analysis of the Impact of Sea-Level Rise and Tropical Cyclones on Mangroves in the Context of Climate Change Liu, S., Zhu, Y., Xiao, H., (...), Liu, D. *Water* (Switzerland), 2024 <https://www.scopus.com/pages/publications/85213289963?origin=scopusAI>
53. Plasticity in the Above- and Below-Ground Development of Mangrove Seedlings in Response to Variation in Soil Bulk Density Ola, A., Staples, T.L., Robinson, N., Lovelock, C.E. *Estuaries and Coasts*, 2020 <https://www.scopus.com/pages/publications/85075612197?origin=scopusAI>
54. Self-purification in tropical upland rivers as affected by environmental factors: the case of Balili River in Benguet, Philippines Ago, M.K., Caasi, R.D., Kege-e, D.S., Napaldet, J.T. *International Journal of Phytoremediation*, 2024 <https://www.scopus.com/pages/publications/85199767569?origin=scopusAI>
55. Coupled 1D-2D tidal inundation modelling of the Sundarbans in the Ganges Delta Karim, F., Nahiduzzaman, S., Ahmmad, R., (...), Alam, R. *Lecture Notes in Civil Engineering*, 2026 <https://www.scopus.com/pages/publications/105031887888?origin=scopusAI>
56. Prolonged coastal inundation detected with synthetic aperture radar significantly retarded functional recovery of mangroves after major hurricanes Yu, M., Gao, Q. *Landscape Ecology*, 2023 <https://www.scopus.com/pages/publications/85140128986?origin=scopusAI>
57. Assessing the effects of using high-quality data and high-resolution models in valuing flood protection services of mangroves Menéndez, P., Losada, I.J., Torres-Ortega, S., (...), Beck, M.W. *PLoS ONE*, 2019 <https://www.scopus.com/pages/publications/85070914954?origin=scopusAI>
58. Flood Risks, Mangrove Ecosystem and Coastal Megacities: Assessing the Adaptation Benefits of Mangrove Ecosystem to Urban Flood Risk in the H/E Ward of Mumbai Tyagi, P. *Indian Journal of Environmental Protection*, 2022 <https://www.scopus.com/pages/publications/85136078405?origin=scopusAI>
59. Mangrove squeeze potential due to combined effect of sea level rise and decadal human land use and changes in a sub-tropical estuarine environment Almeida, L.P., Cruz, L., Pinto, M.W., (...), Klein, A.H.D.F. *Regional Studies in Marine Science*, 2025 <https://www.scopus.com/pages/publications/105011540903?origin=scopusAI>
60. Assessment and mitigation of tangible flood damages driven by climate change in a tropical city: Hat Yai Municipality, southern Thailand Tabucanon, A.S., Kurisu, K., Hanaki, K. *Science of the Total Environment*, 2021 <https://www.scopus.com/pages/publications/85107673726?origin=scopusAI>